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5.1.3 What you will learn

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This week is all about solving nonsingular linear systems via LU (with or without pivoting) and Cholesky factorization. In practice, solving $Ax = b$ is *not* accomplished by forming the inverse explicitly and then computing $x = A^{-1}b$. Instead, the matrix A is factored into the product of triangular matrices and it is these triangular matrices that are employed to solve the system. This requires fewer computations.

Upon completion of this week, you should be able to

- Link Gaussian elimination to LU factorization.
- View LU factorization in different ways: as Gaussian elimination, as the application of a sequence of Gauss transforms, and the operation that computes L and U such that $A = LU$.
- State and prove necessary conditions for the existence of the LU factorization.
- Extend the ideas behind Gaussian elimination and LU factorization to include pivoting.
- Derive different algorithms for LU factorization and for solving the resulting triangular systems.
- Employ the LU factorization, with or without pivoting, to solve $Ax = b$.
- Identify, prove, and apply properties of Hermitian Positive Definite matrices.
- State and prove conditions related to the existence of the Cholesky factorization.
- Derive Cholesky factorization algorithms.
- Analyze the cost of the different factorization algorithms and related algorithms for solving triangular systems.

Discussion

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